FRDM-IMXRT700

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Revision History

Rev. Code	Date	By	Description
A	2026-04-08	James Fan	Initialize release

Jumper Setting

REF DES	JUMPER(DEFAULT)	PAGE NAME
J31	1-2	04 System Power 1
J33	1-2	05 System Power 2
J2	5-6	05 System Power 2
JP1,JP3,JP4,JP5,JP7,JP8,JP9, JP10,JP11,JP19,JP21,JP26, JP61,JP62,JP63,JP70	1-2	06 RT700 POWER
JP2	2-3	06 RT700 POWER
JP43,JP58	Open	12 MIPI Display
JP16	1-2	13 Arduino/Mikroe/User Button
JP44	1-2	14 I3C/DMIC/Debug/TRACE
JP30,JP67	OPEN	15 Audio Codec
JP15,JP54,JP55,JP56	1-2	17 Boot/SYSTEM RESET
JP14,JP23	1-2	18 MCU Link
JP6,JP12,JP18,JP20,JP24,JP27	Open	18 MCU Link

Switch Setting

REF DES	SWITCH(DEFAULT)	PAGE NAME
SW10	on,off	17 Boot/SYSTEM RESET

XSPI Usage

XSPI0 w/ Cache QSPI Nor Flash
XSPI1 w/ Cache 16-bit HYPERRAM and QSPI Nand Flash
XSPI2 w/o Cache 16-bit HYPERRAM (Graphics/Data)

1. Unless Otherwise Specified:

   All resistors are in ohms, 1/16 Watt,0402
   All capacitors are in uF,0402
   All voltages are DC
   All polarized capacitors are aluminum electrolytic

2. Interrupted lines coded with the same letter or letter combinations are electrically connected.

3. Device type number is for reference only. The number varies with the manufacturer.

4. Special signal usage:
   _B Denotes - Active-Low Signal
   <> or [] Denotes - Vectored Signals

5. Interpret diagram in accordance with American National Standards Institute specifications, current revision, with the exception of logic block symbology.

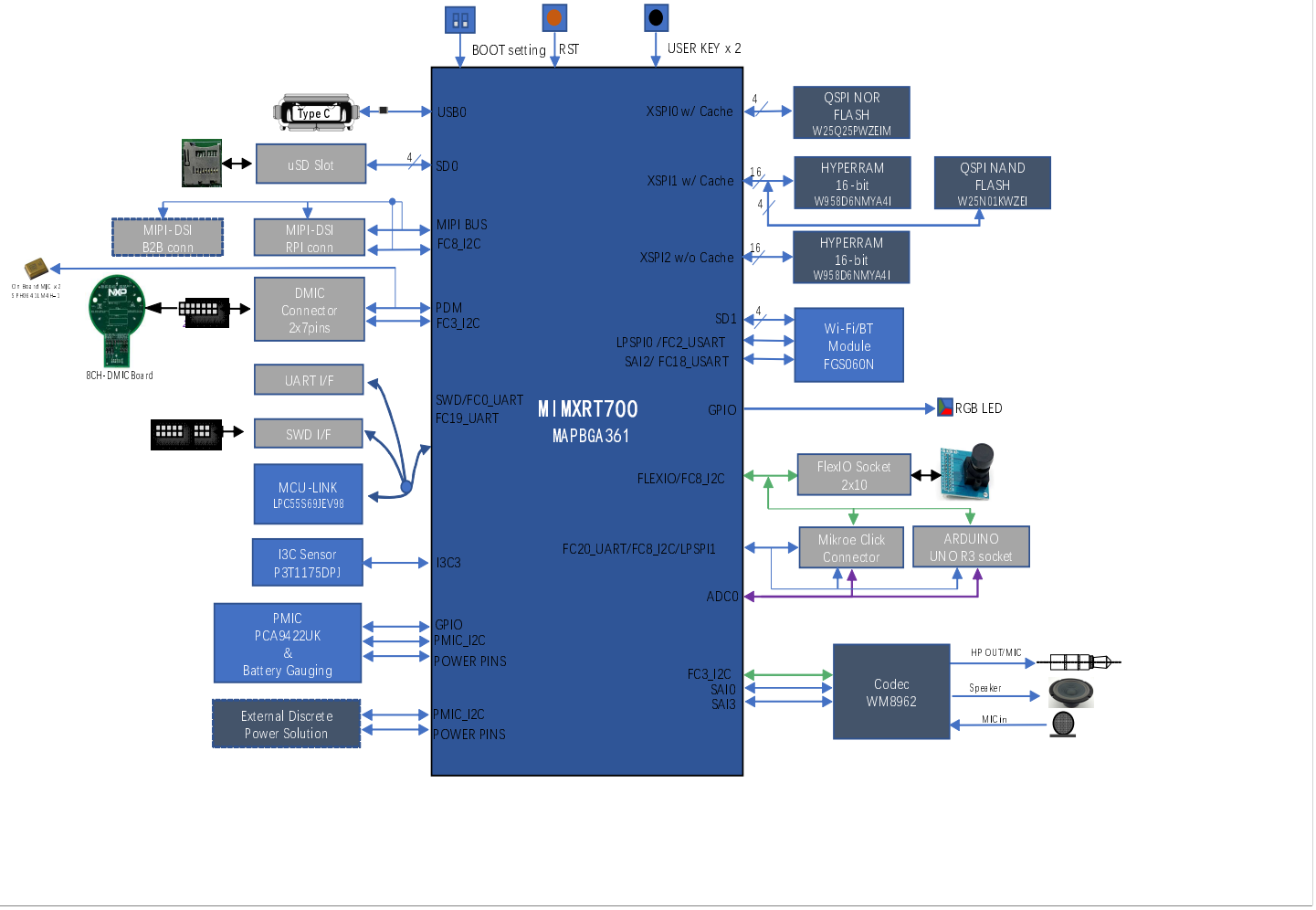
Microcontroller Product Group
6501 William Cannon Drive West
Austin, TX 78735-8598

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ICAP Classification: CP: IUO: PUBI:

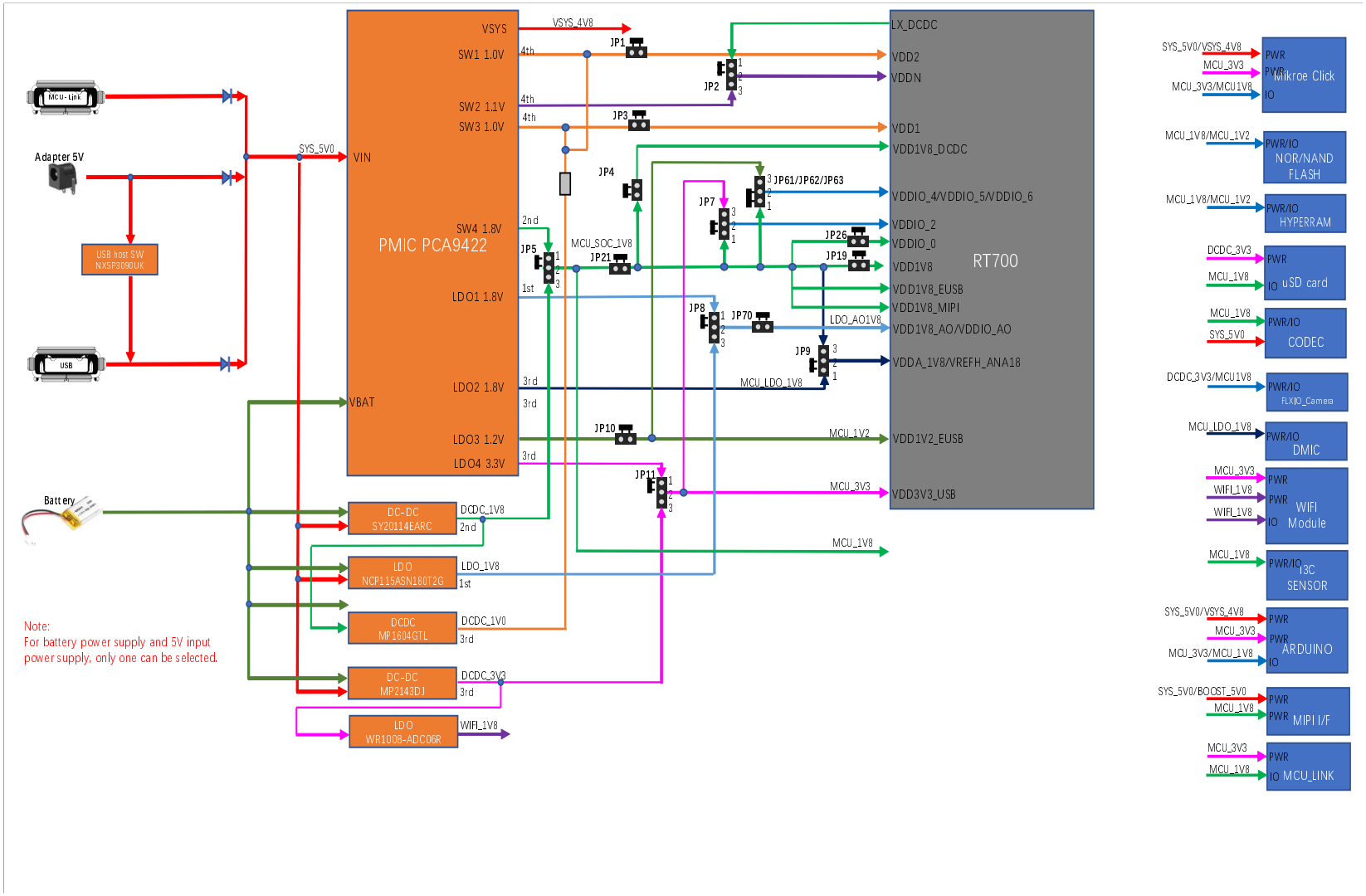
Designer: James Fan	Drawing Title: FRDM-IMXRT700		
Drawn by: James Fan	Page Title: 01 Cover		
Approved: Yes	Size C	Document Number SCH-96317, PDF: SPF-96317,	Rev A
	Date: Thursday, April 09, 2026	Sheet 1 of 20	

FRDM-IMXRT700 BLOCK DIAGRAM



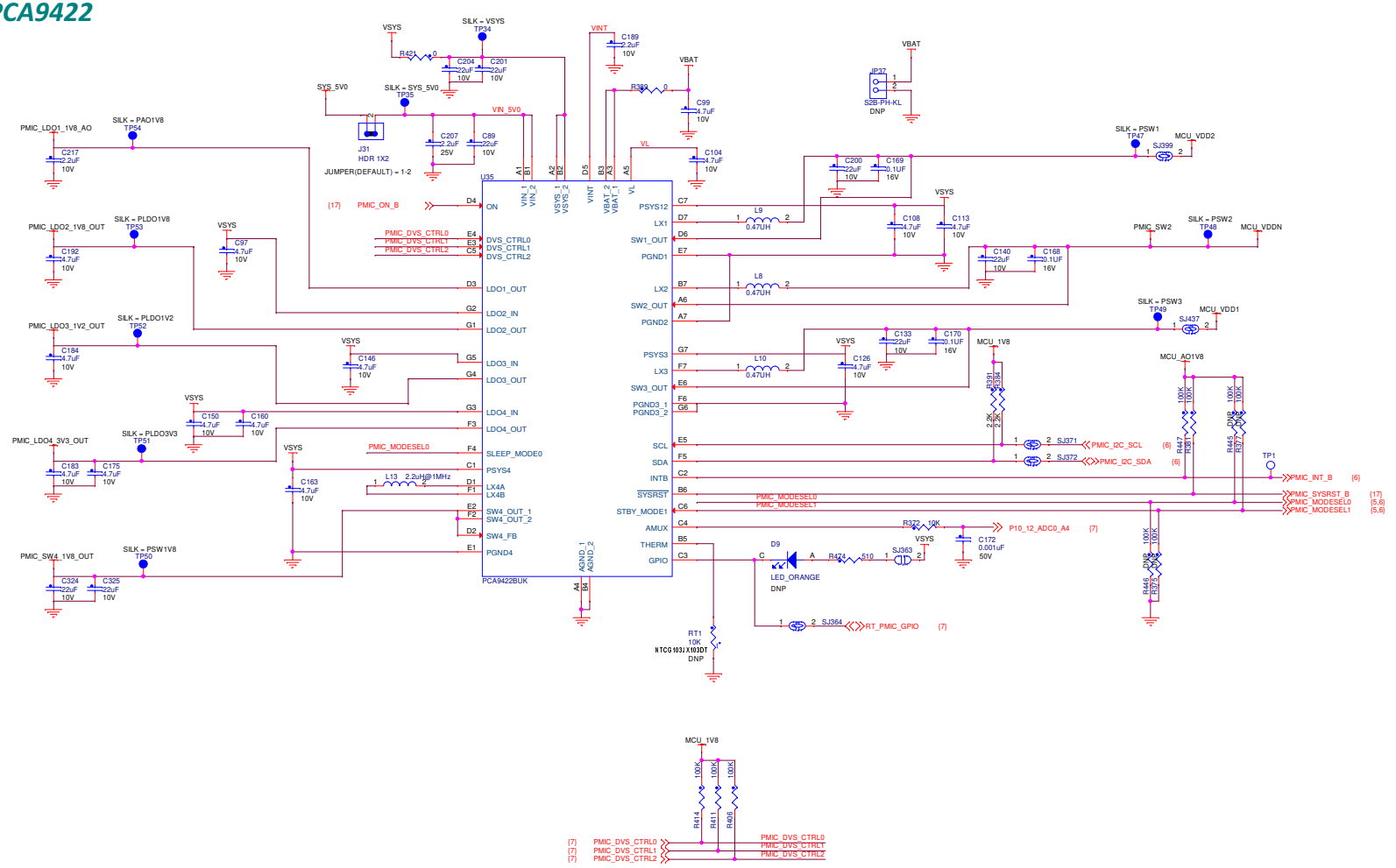
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Drawing Title:	FRDM-IMXRT700	
Page Title:	02 FRDM Block Diagram	
Size: C	Document Number: SCH-96317, PDF: SPF-96317,	Rev: A
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FRDM-IMXRT700 POWER SUPPLY



ICAP Classification:	CP:	IUO: PUBI:
Drawing Title:	FRDM-IMXRT700	
Page Title:	03 FRDM POWER SUPPLY	
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PMIC PCA9422



ICAP Classification:	Classification:	IUO:	PUBL:
Drawing Title:	FRDM-IMXRT700		
Page Title:	04 System Power 1		
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5V Adapter

Reverse Voltage Protection

Barrel Connector

Over-Voltage Protection

SW5(2-3/5-6) ->5V OFF
SW5(1-2/4-5) ->5V ON

DCDC3V3

Current 3A
I_Quiescent : 40μA

MODESEL0	MODESEL1	GPIO_MODESEL3	DCDC_Vout
0	0	0	->1.0V
0	0	1	->1.1V-0.7V
0	1	0	->0.63V
1	0	0	->0.5V

DCDC1V8

LDO1V8

Board Mounting Holes

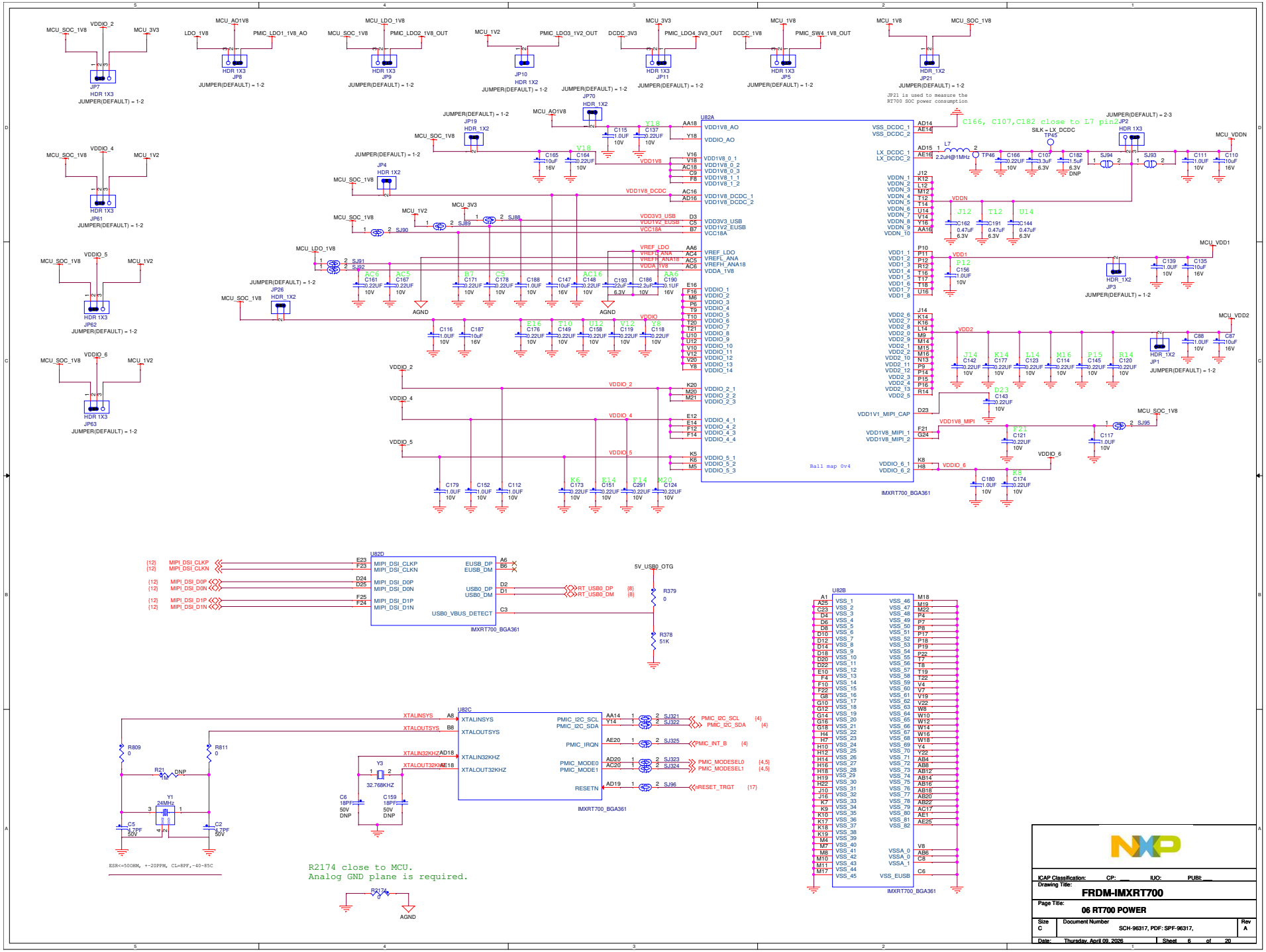
H1 MOUNT_HOLE_125 H2 MOUNT_HOLE_125 H3 MOUNT_HOLE_125 H4 MOUNT_HOLE_125

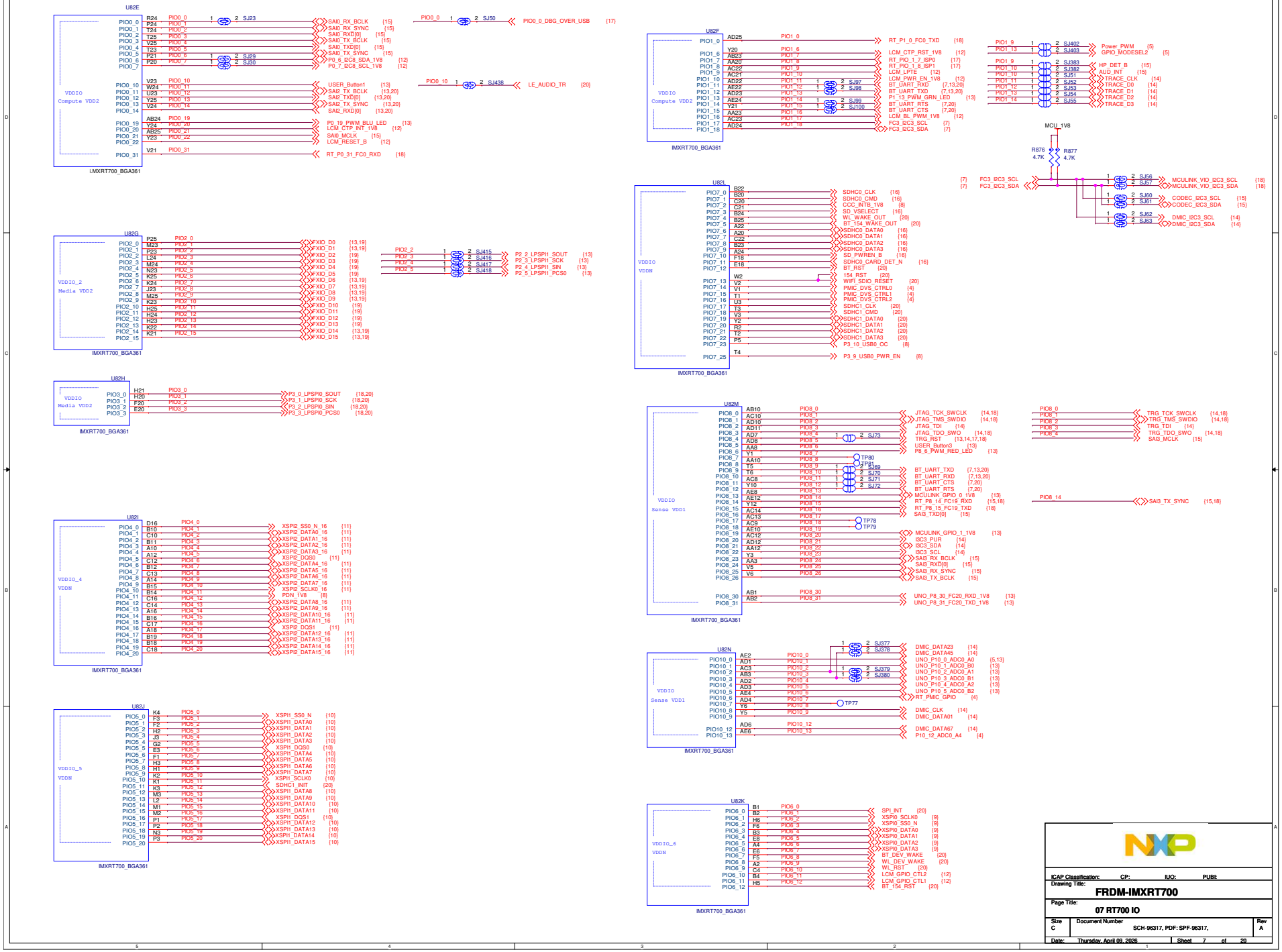
Ground TPs

Layout Note: Place Ground TPs to assist signal measurement.

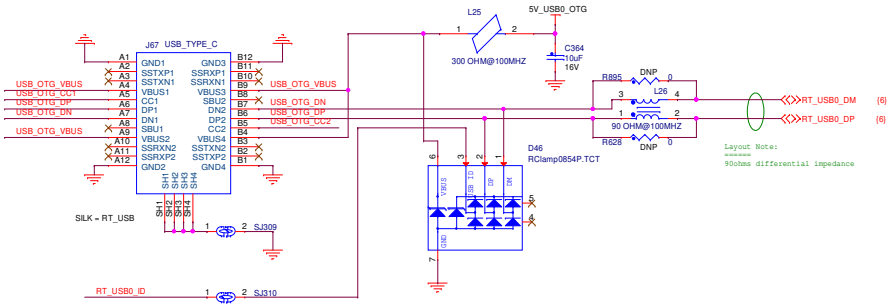
TP6 SILK - GND DNP

ICAP Classification:	CP: ___ IUO: PUBI: ___
Drawing Title:	FRDM-IMXRT700
Page Title:	05 System Power 2
Size: C	Document Number: SCH-96317, PDF: SPF-96317, Rev: A
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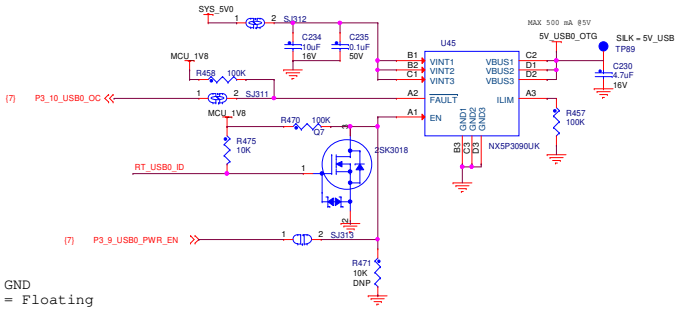

C166, C107,C182 close to L7 pin2.
R2174 close to MCU.
Analog GND plane is required.
IMXRT700_BGA361
FRDM-IMXRT700
06 RT700 POWER
SCH-96317, PDF: SPF-96317,
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U82E
i.MXRT700_BGA361
U82F
IMXRT700_BGA361
U82G
IMXRT700_BGA361
U82L
IMXRT700_BGA361
U82H
IMXRT700_BGA361
U82M
IMXRT700_BGA361
U82I
IMXRT700_BGA361
U82N
IMXRT700_BGA361
U82J
IMXRT700_BGA361
U82K
IMXRT700_BGA361
MCU_1V8
R876 4.7K
R877 4.7K
ICAP Classification:
CP:
IUO:
PUB:
Drawing Title:
FRDM-IMXRT700
Page Title:
07 RT700 IO
Size C
Document Number
SCH-96317, PDF: SPF-96317,
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RT700 USB0 HOST/DEVICE

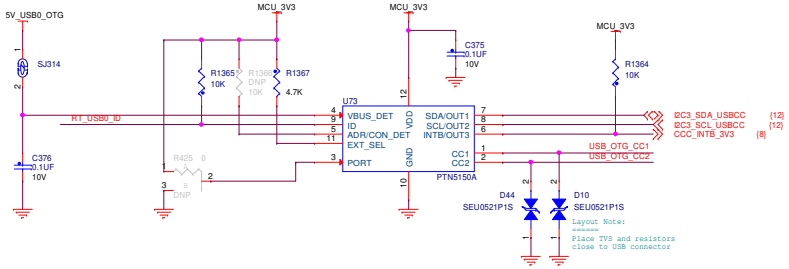


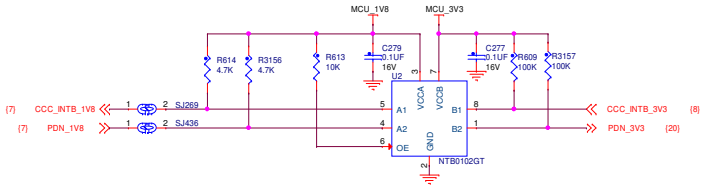
USB0 Power



Host --> ID = GND
Device --> ID = Floating

CC Logic



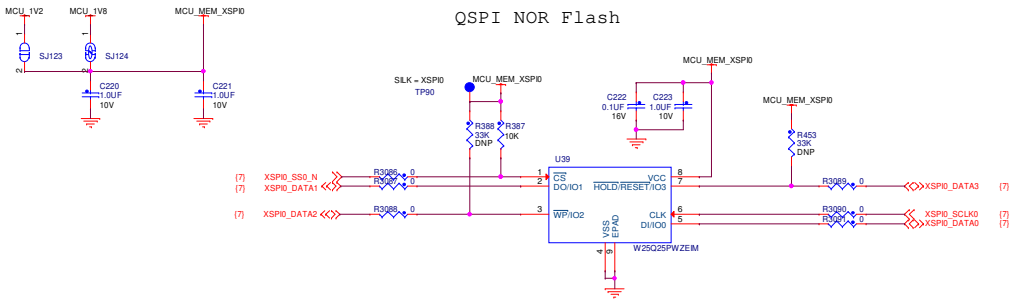


EXT_SEL	PORT	ADR/CON_DET
0---CC2 1---CC1	PORT=1: DFP mode PORT=0: UFP mode PORT=Float: DRP mode	CON_DET=1, Connection Detected, I2C Address: 0x7A CON_DET=0, No Connection, I2C Address: 0x3A
Default: 1	Default: PORT=Float: DRP mode	Default: Float, no I2C function

ICAP Classification:	CP:	IUO:	PUBI:
Drawing Title:	FRDM-IMXRT700		
Page Title:	08 USB Power Supply		
Size: C	Document Number: SCH-96317, PDF: SPF-96317,		Rev: A
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XSPI0 w/ Cache 4-bit FLASH

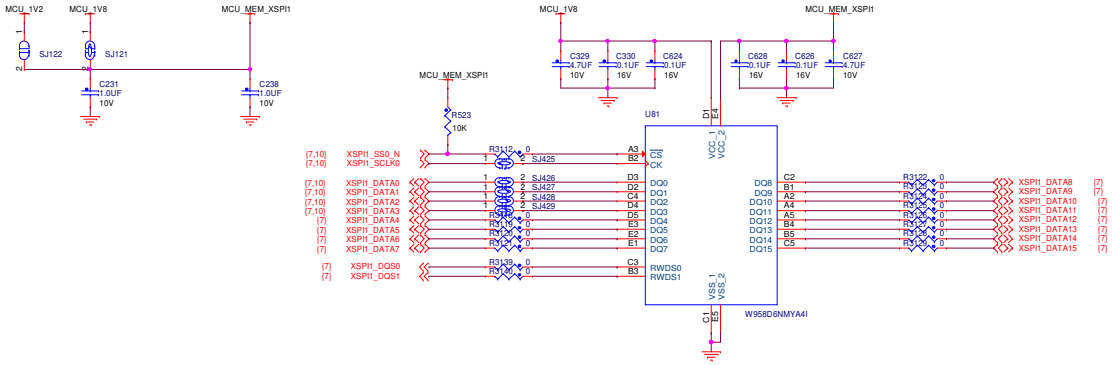
Notes: If evaluating 1.2V memory, solder SJ123 DNP SJ124 the voltage of VDDIO_6/JP63 also needs to be switched to 1.2V.



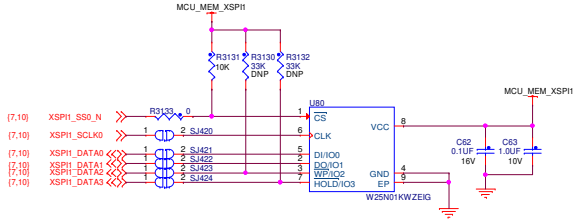
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Drawing Title:	FRDM-IMXRT700
Page Title:	09 XSPI0 Flash
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XSPI1 w/ Cache 16-bit HYPERRAM and QSPI NAND Flash

Notes: If evaluating 1.2V memory, solder SJ122, DNP SJ121, the voltage of VDDIO_5/JP62 also needs to be switched to 1.2V.



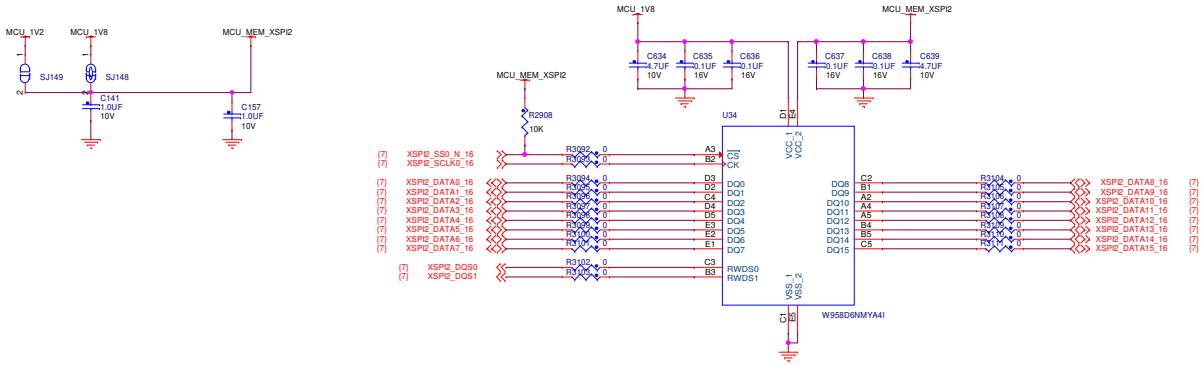
QSPI NAND Flash



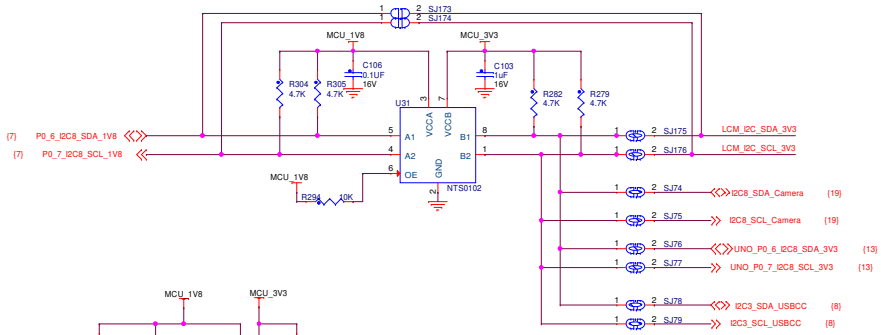
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Page Title:	10 XSPI1 HYPERRAM		
Size: C	Document Number: SCH-96317, PDF: SPF-96317,		Rev: A
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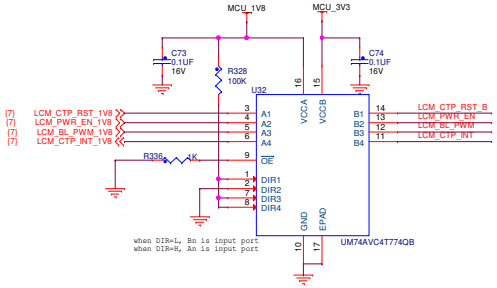
XSPI2 w/o Cache 16-bit HYPERRAM (Graphics/Data)

Notes: If evaluating 1.2V memory, solder SJ149 , DNP SJ148, the voltage of VDDIO_4/JP61 also needs to be switched to 1.2V.

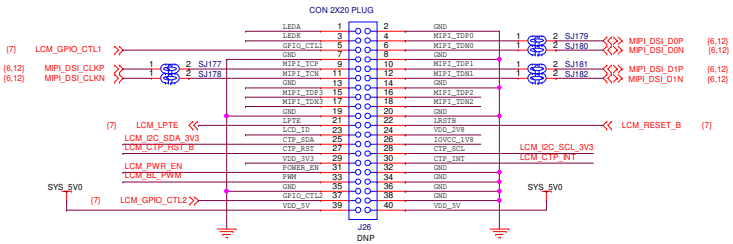


ICAP Classification:	CP: ___ IUO: PUBI: ___	
Drawing Title:	FRDM-IMXRT700	
Page Title:	11 XSPI2 HYPERRAM	
Size C	Document Number SCH-96317, PDF: SPF-96317,	Rev A
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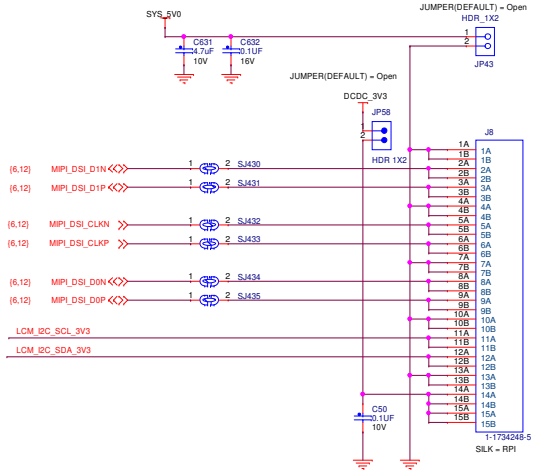




ZC143AC72MIPI Display Interface

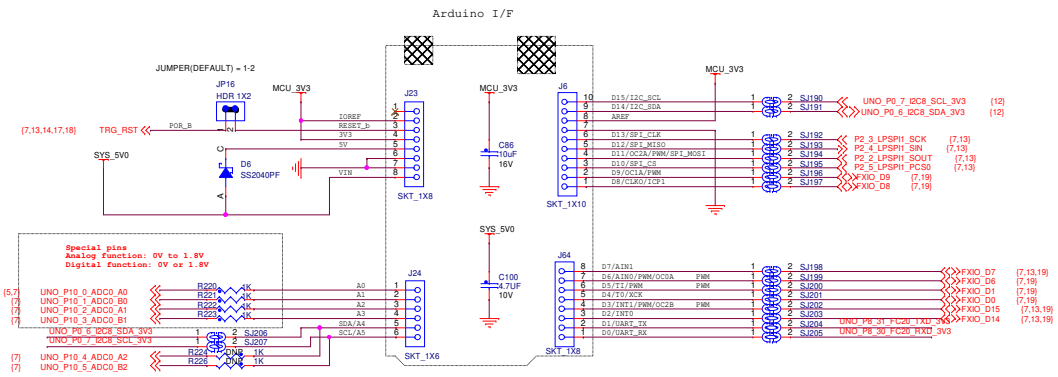


RPI Display



ICAP Classification:	CP:	IUO: PUBI:
Drawing Title:	FRDM-IMXRT700	
Page Title:	12 MIPI DSI	
Size: C	Document Number: SCH-96317, PDF: SPF-96317,	Rev: A
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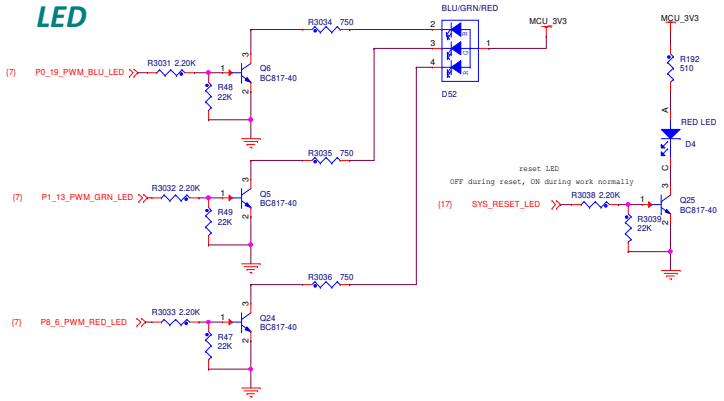
Arduino Connector

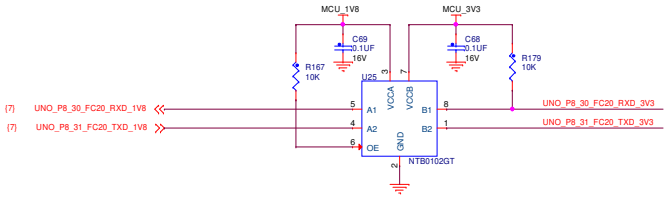


The connectors used are 2.54mm sockets

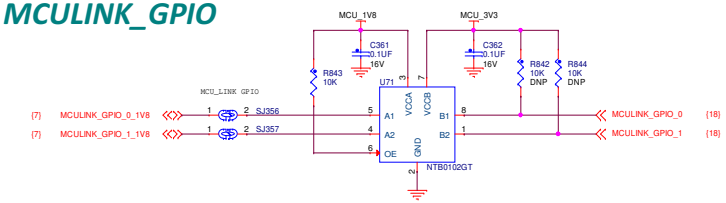
NOTE: Arduino shield used should not exceed 200 mA power requirements through 3V3.

LED



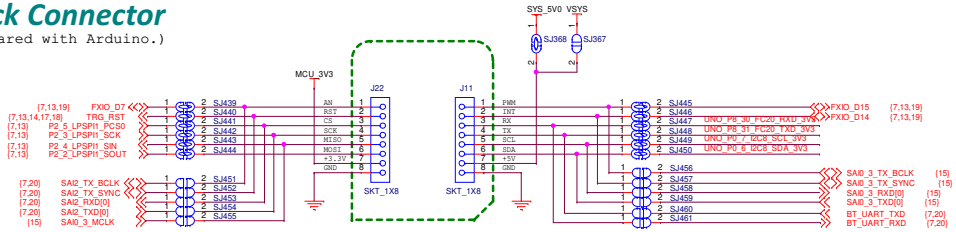


MCULINK_GPIO

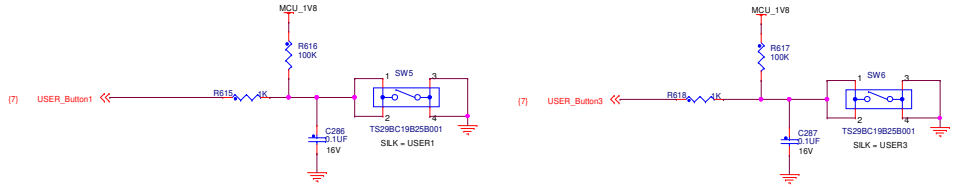


Mikroe Click Connector

(Signals are shared with Arduino.)

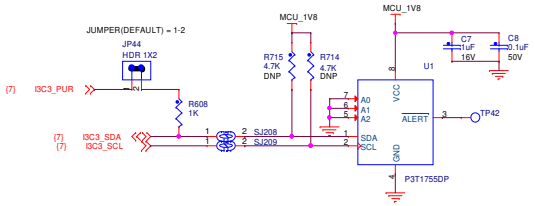


User Button



ICAP Classification:	CP:	IUO:	PUBI:
Drawing Title:	FRDM-IMXRT700		
Page Title:	13 Arduino/Mikroe/User Button		
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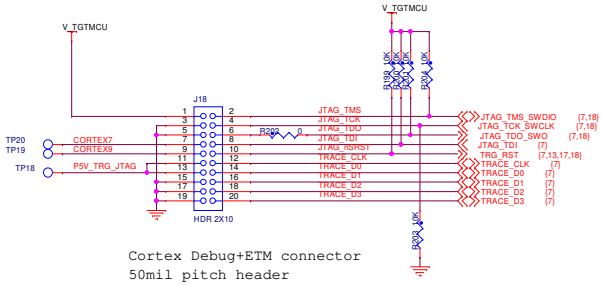
I3C SENSOR



I2C ADDRESS: 1001000

DEBUG&TRACE

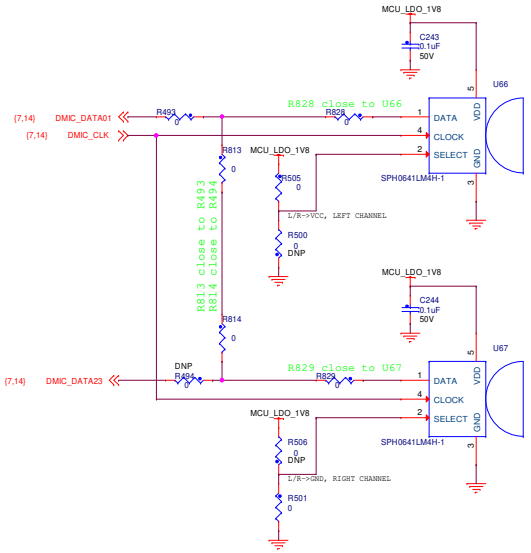
Short JP18 when using J18 to debug MCU



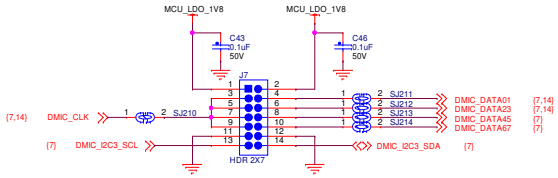
Cortex Debug+ETM connector
50mil pitch header

OnBoard DMIC

Notes: Placing the mic under PCB which is opening to face the user

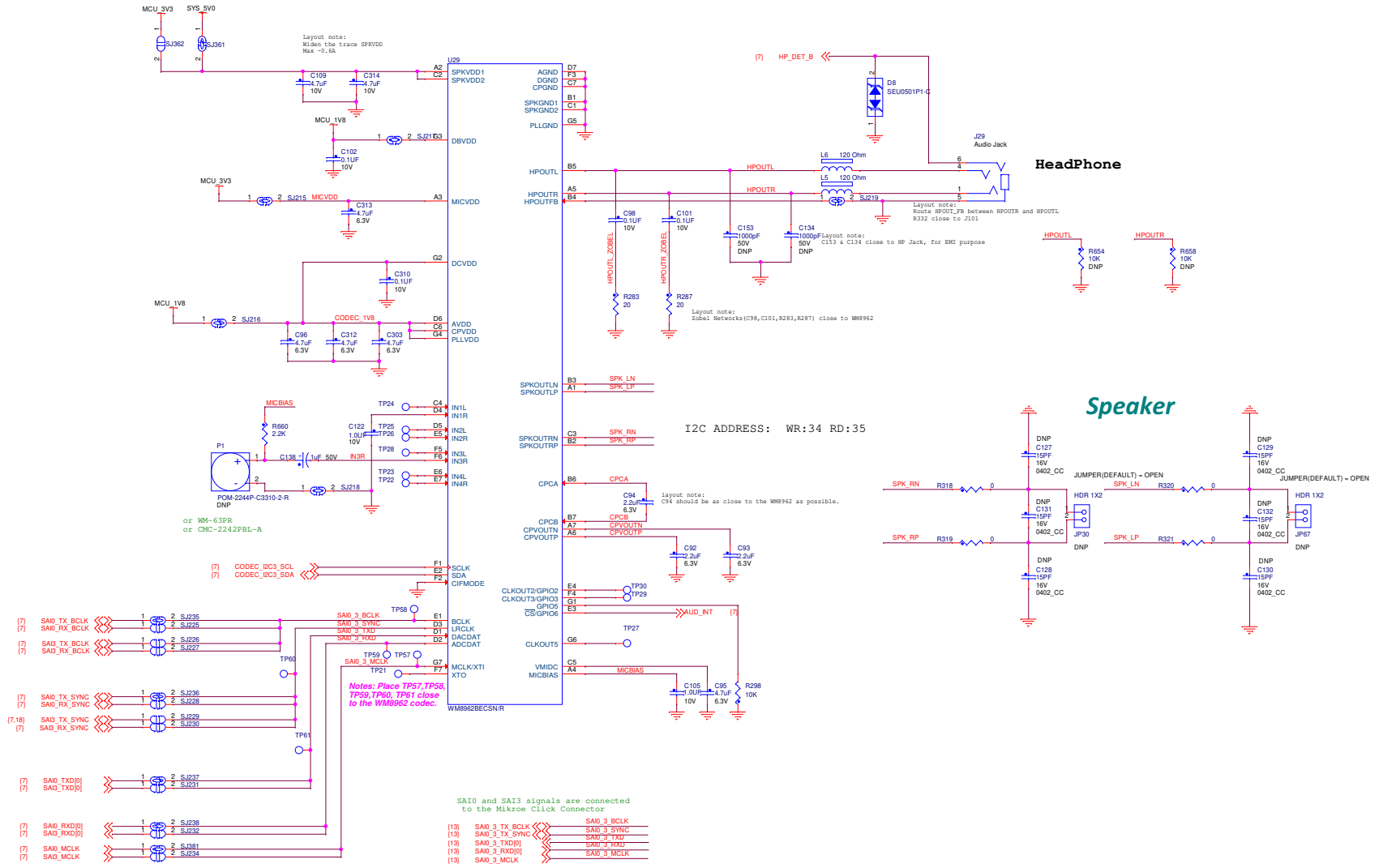


8CH-DMIC Extension



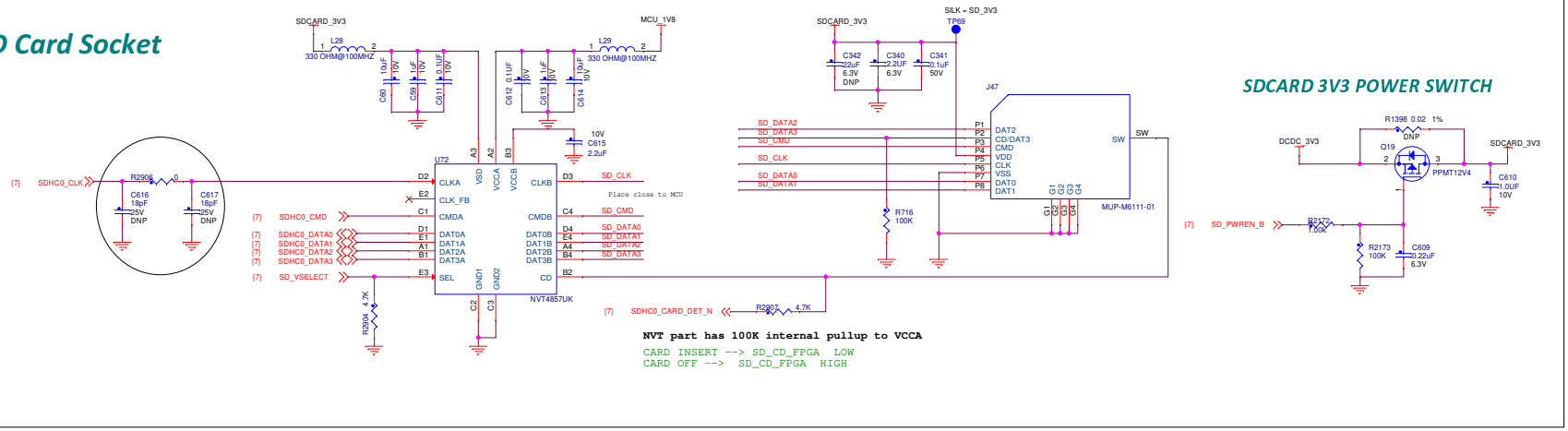
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Drawing Title:	FRDM-IMXRT700	
Page Title:	14 I3C/DMIC/Debug/TRACE	
Size: C	Document Number: SCH-96317, PDF: SPF-96317,	Rev: A
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Audio Codec WM8962, CPU0 SAI1 or CPU1 SAI3 is connected to WM8962 codec.



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Drawing Title:	FRDM-IMXRT700		
Page Title:	15 Audio Codec		
Size: C	Document Number: SCH-96317, PDF: SPF-96317,		Rev: A
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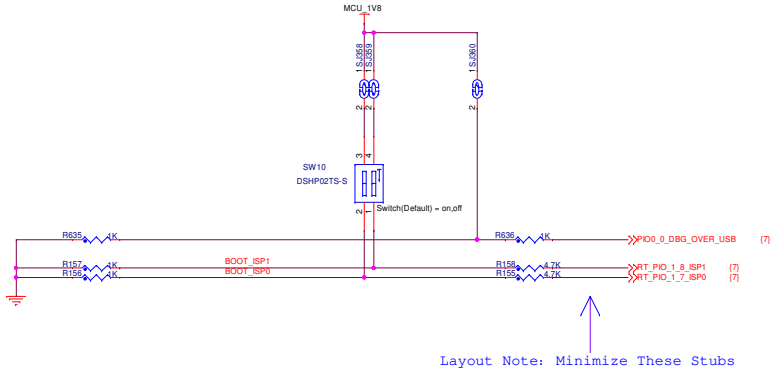
SD Card Socket



SDCARD 3V3 POWER SWITCH

ICAP Classification:	CP: ___ IUO: PUBI: ___	
Drawing Title:	FRDM-IMXRT700	
Page Title:	16 SD Card/M.2 Socket	
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External Boot Switch

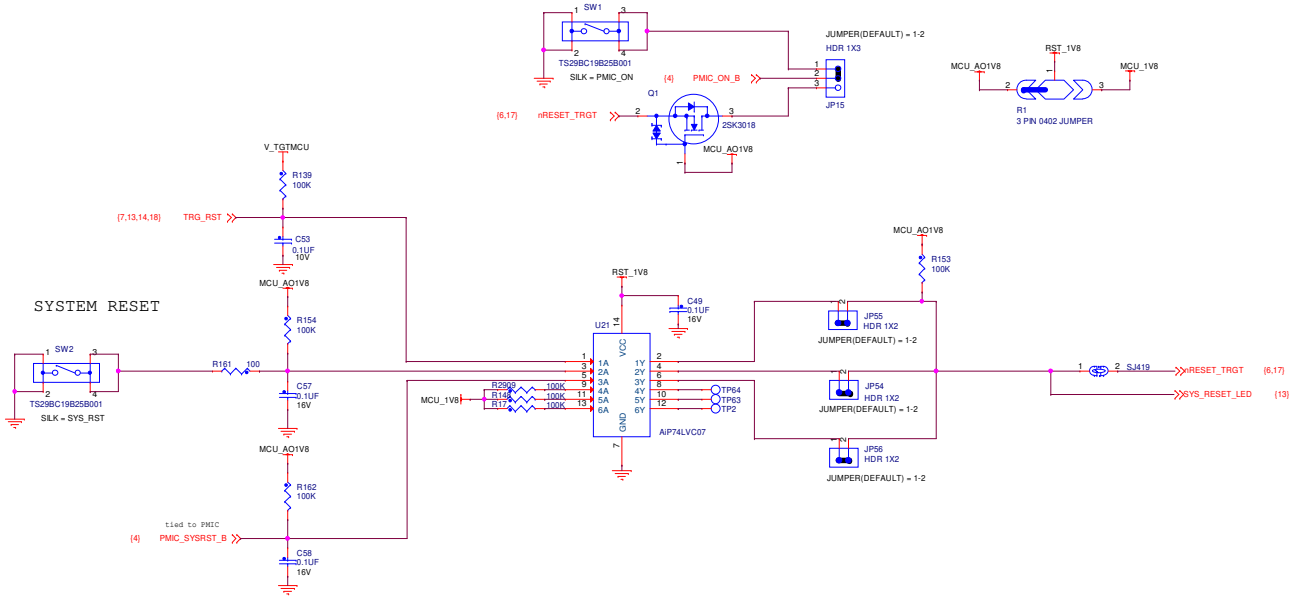


Boot Mode pin settings

BOOT_ISP[1:0]	Boot Type
00 (SW10-x00)	SDHC0 eMMC
01 (SW10-x01)	XSPI0
10 (SW10-x10)	Auto ISP (UART, SPI-slave, I2C-slave, USB-HID, eUSB)
11 (SW10-x11)	XSPI1

DBG_OVER_USB	Function Enablement
0 (PIO0_0)	Enable
1 (PIO0_0)	Disable

System Reset Logic



ICAP Classification: CP: ___ IUO: PUBI: ___

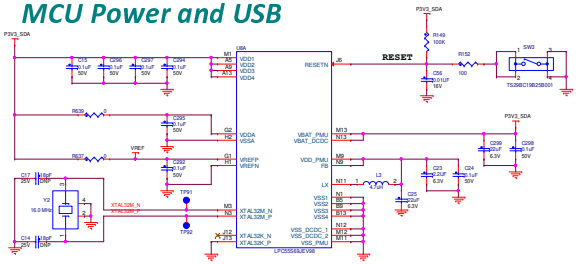
Drawing Title: **FRDM-IMXRT700**

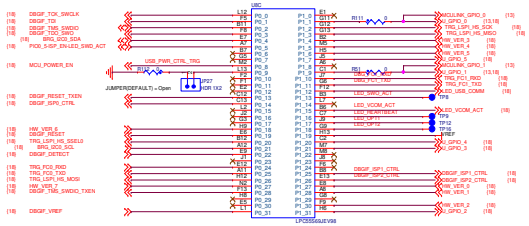
Page Title: **17 Boot/SYSTEM RESET**

Size	Document Number	Rev
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MCU Power and USB

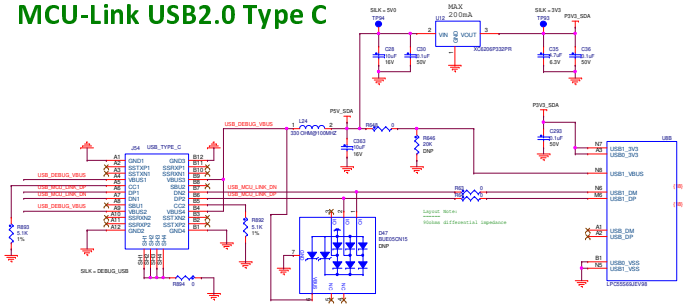


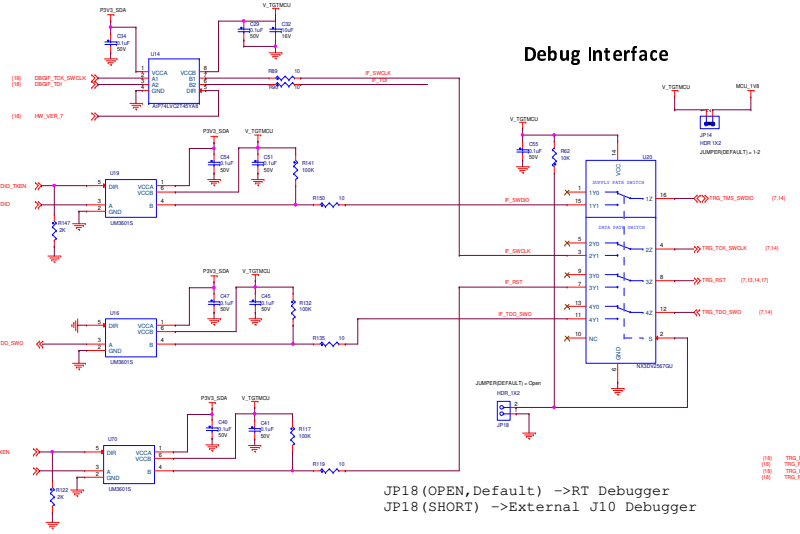


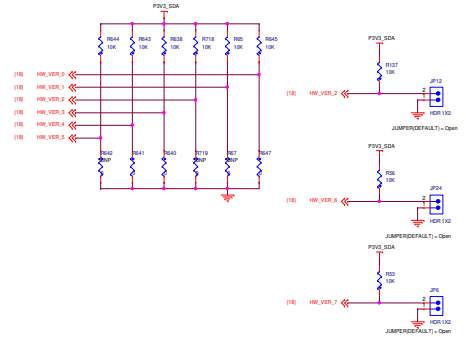
Items	Description	OB setting
HW_VER_0(PIO1_27)	Power negotiation when low	0
HW_VER_1(PIO1_28)	Reserved	1
HW_VER_2(PIO1_30)	USBSIO bridge enabled / not present when 0	1
HW_VER_3(PIO1_4)	OB (0) / Pro (1) type select	0
HW_VER_4(PIO1_5)	Board identity code used (when low)	0
HW_VER_5(PIO1_6)	Power measurement enabled/present when low	1
HW_VER_6(PIO0_18)	VCOM disabled when low	1
HW_VER_7(PIO0_27)	SWD debug disabled when low	1

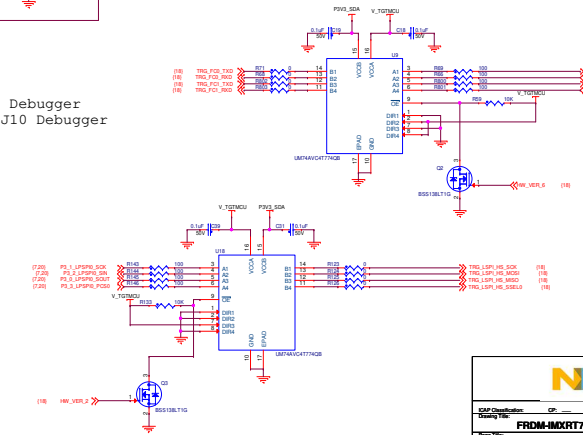


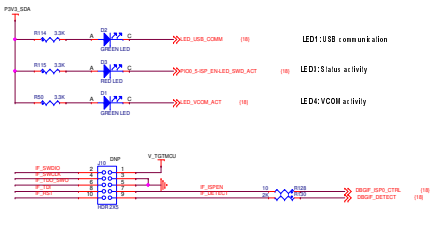
MCU-Link USB2.0 Type C

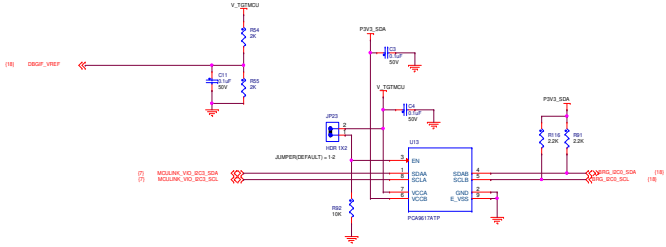








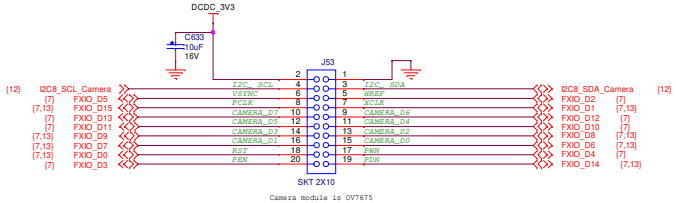








FLEXIO Camera Connector



Camera module is OV7675

ICAP Classification: CP: IUO: PUBI:

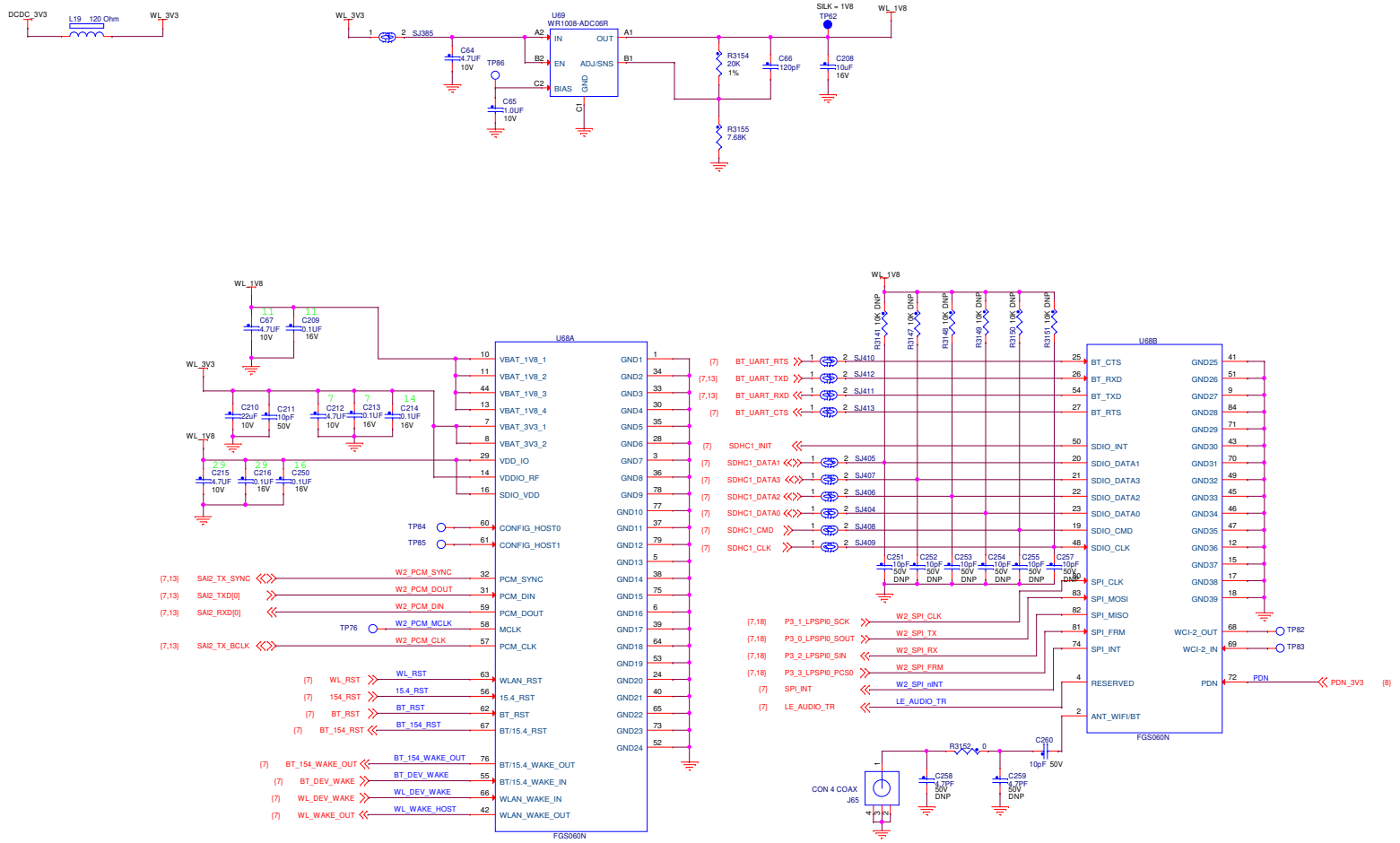
Drawing Title: **FRDM-IMXRT700**

Page Title: **19 FLEXIO Camera**

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C	SCH-96317, PDF: SPF-96317,	A

Date: Thursday, April 09, 2026 Sheet 19 of 20

WiFi/BT Module FGS060N



ICAP Classification:	CP:	IUO:	PUB:
Drawing Title:	FRDM-IMXRT700		
Page Title:	20 WIFI/BT Module		
Size: C	Document Number: SCH-96317, PDF: SPF-96317,		Rev: A
Date: Thursday, April 09, 2026	Sheet 20 of 20		